

PRESS RELEASE

EXTRACTING ENERGY AND HEAT FROM THE VACUUM

Quantum theory tells us that empty space is not truly empty, but rather contains an enormous amount of untapped electromagnetic energy known as the **zero-point energy**, or ZPE. (The adjective "zero-point" signifies that such energy exists even at a temperature of absolute zero where no thermal effects remain.) Such energy can be traced to radiation from the fluctuating quantum motion of charged particles distributed throughout the universe (see "Where does the zero-point energy come from?", *New Scientist*, December 1989). The ZPE is responsible for such diverse phenomena as the van der Waals forces of chemical attraction at absolute zero, the perturbation of atomic spectral lines known as the Lamb Shift, and the Casimir effect, a unique attractive quantum force between closely-spaced metal plates.

The energy associated with the ZPE is known to be essentially inexhaustible and ubiquitous, so a question that arises in the mind of the technologist is whether such energy can be "mined" for practical use - and thereby constitute the "Holy Grail" of energy research. Although deferring for the present any technological considerations for actual application and use, Daniel Cole of IBM in Essex Junction, Vermont, and Harold Puthoff of the Institute for Advanced Studies in Austin, Texas, have addressed this issue in a recent article entitled "Extracting energy and heat from the vacuum." Writing in *Physical Review*, these researchers critically examine the underlying energetic and thermodynamic processes involved in such a concept and come to the conclusion that, yes, in principle, proposals to extract energy from vacuum fluctuations do not violate any fundamental precepts (*Physical Review E*, vol 48, p 1562).

What has been shown is that the basic concept of the conversion of vacuum energy to other potentially useful forms is a legitimate and viable physics principle. What remains, however, as with solar and thermonuclear energy, is the matter of engineering and demonstration as to whether vacuum energy conversion can be developed to the point that it constitutes a significant energy resource.

Steve -
You are right
about the potential
of zero-point-energy
technologies
Hal Puthoff